

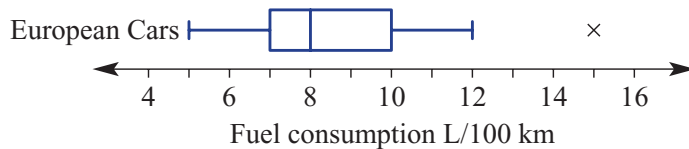
5.5 Boxplots



The five-figure summary (Min, Q_1 , Q_2 , Q_3 , Max) can be represented in graphical form as a boxplot. Boxplots are graphs that summarise single data sets. They clearly display the minimum and maximum values, the median, the quartiles and any outliers. Q_1 , Q_2 and Q_3 divide the data into quarters. Boxplots also give a clear indication of how data is spread, as the IQR (interquartile range) is shown by the width of the central box.

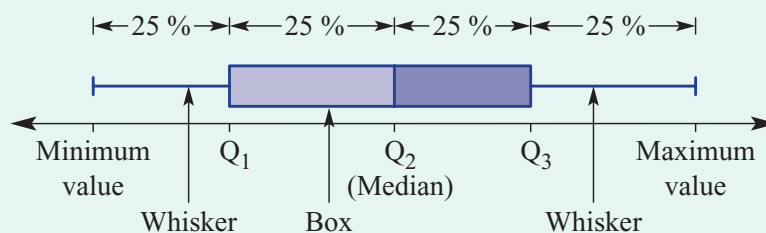
► Let's start: Fuel consumption

This boxplot summarises the average fuel consumption (litres per 100 km) for a group of European-made cars.



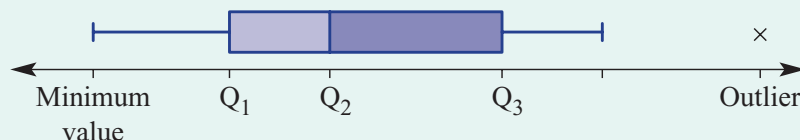
- What does each part of the boxplot represent in terms of the five-figure summary?
- What do you think the cross (x) represents?
- Describe how you can use the boxplot to find the IQR.
- What would you expect the fuel consumption to be above for the top 25% of cars?

- A **boxplot** (also called a box-and-whisker plot) can be used to summarise a data set. It displays the five figure summary (Min, Q_1 , Q_2 , Q_3 , Max), as shown.

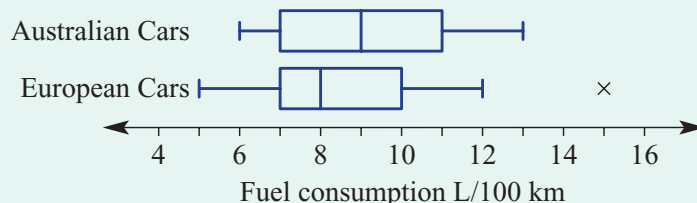


Boxplot A diagram using rectangles and ranges to show the spread of a set of data, using five important values

- An outlier is marked with a cross (x).
 - An outlier is greater than $Q_3 + 1.5 \times \text{IQR}$ or less than $Q_1 - 1.5 \times \text{IQR}$.
 - $\text{IQR} = Q_3 - Q_1$.
 - The whiskers stretch to the lowest and highest data values that are not outliers.



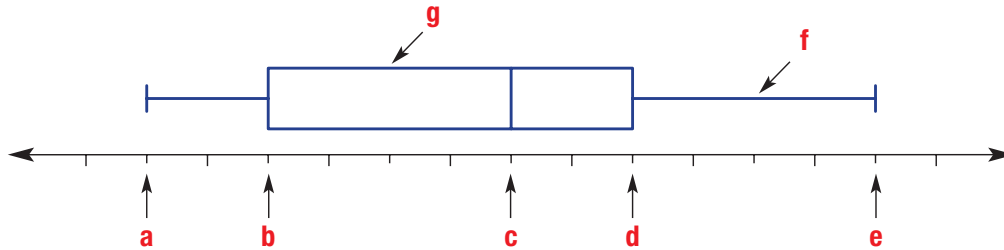
- Parallel boxplots are boxplots drawn on the same scale. They are used to compare data sets within the same context.



Exercise 5E

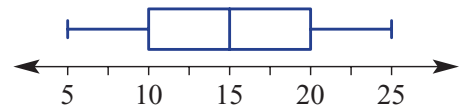
Understanding

- 1 Label the parts of the boxplot below.



- 2 For this simple boxplot, find:

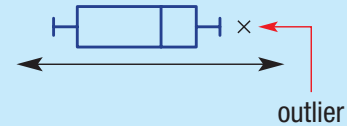
- a** the median (Q_2)
- b** the minimum
- c** the maximum
- d** the range
- e** the lower quartile (Q_1)
- f** the upper quartile (Q_3)
- g** the interquartile range (IQR)



- 3 Construct a boxplot showing these features.

- a** Min = 1, $Q_1 = 3$, $Q_2 = 4$, $Q_3 = 7$, max = 8
- b** Outlier = 5, minimum above outlier = 10, $Q_1 = 12$, $Q_2 = 14$, $Q_3 = 15$, max = 17

Boxplot shape:



Include an even scale.

- 4 Select from the list below to fill in the blanks.

Minimum, Q_1 , Q_2 , Q_3 , Maximum

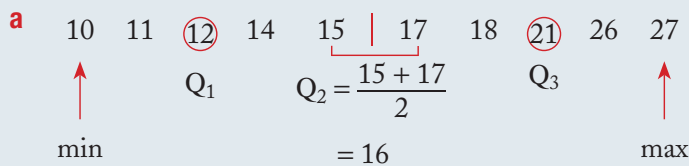
- a** The top 25% of data is above _____.
- b** The middle 50% of data is between ____ and ____.
- c** The lowest or first 25% of data is between the ____ and ____.
- d** The highest or last 25% of data is between ____ and the ____.

Example 12 Constructing boxplots with no outliers

Consider the given data set.

12 26 14 11 15 10 18 17 21 27

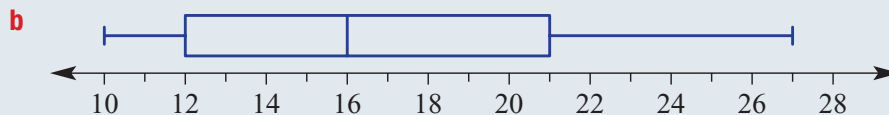
- a** Find the five-figure summary (the minimum, lower quartile (Q_1), median (Q_2), upper quartile (Q_3) and the maximum).
b Draw a boxplot to summarise the data.

Solution

Minimum = 10

$Q_1 = 12$, $Q_2 = 16$, $Q_3 = 21$

Maximum = 27

**Explanation**

- a** Order the data from smallest to largest.
 Locate the median (Q_2) first. For 10 data values there are two middle values, 15 and 17.
 Average these to find Q_2 .
 Split the data in half at the median. Q_1 is the middle value of the lower half. Q_3 is the middle value of the upper half. The minimum is the smallest value and the maximum the largest value.
- b** Draw an even scale covering the minimum and maximum values.
 Mark the minimum (10), Q_1 (12), Q_2 (16), Q_3 (21) and the maximum (27) to draw the boxplot.

5 For the given data sets:

- i** find the five-figure summary (minimum, Q_1 , Q_2 , Q_3 , maximum)
ii draw a boxplot to summarise the data

a 11, 15, 18, 17, 1, 2, 8, 12, 19, 15

b 0, 1, 5, 4, 4, 4, 2, 3, 3, 1, 4, 3

c 124, 118, 119, 117, 120, 120, 121, 118, 122

d 62, 85, 20, 34, 40, 66, 47, 82, 25, 32, 28, 49, 41, 30, 22

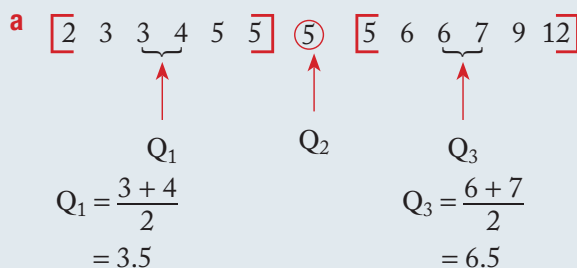
Example 13 Constructing boxplots with outliers

Consider the given data set.

5, 9, 4, 3, 5, 6, 6, 5, 7, 12, 2, 3, 5

- a** Determine the quartiles Q_1 , Q_2 and Q_3 .
b Determine whether any outliers exist.
c Draw a boxplot to summarise the data, marking outliers if they exist.

Solution



b

$$\begin{aligned} \text{IQR} &= 6.5 - 3.5 = 3 \\ Q_1 - 1.5 \times \text{IQR} &= 3.5 - 1.5 \times 3 \\ &= -1 \\ Q_3 + 1.5 \times \text{IQR} &= 6.5 + 1.5 \times 3 \\ &= 11 \end{aligned}$$

$\therefore 12$ is an outlier.



Explanation

Order the data to help find the quartiles.

Locate the median Q_2 (the middle value), then split the data in half above and below this value.

Q_1 is the middle value of the lower half and Q_3 the middle value of the upper half. Average the two middle values to find the median.

Determine $\text{IQR} = Q_3 - Q_1$.

Check for any outliers; i.e. numbers below $Q_1 - 1.5 \times \text{IQR}$ or above $Q_3 + 1.5 \times \text{IQR}$.

There is no data below -1 but $12 > 11$.

Draw a line and mark in a uniform scale reaching from 2 to 12. Sketch the boxplot by marking the minimum 2 and the outlier 12, and Q_1 , Q_2 and Q_3 . The end of the five-point summary is the nearest value below 11; i.e. 9.

6 Consider the data sets below.

- Determine the quartiles Q_1 , Q_2 and Q_3 .
- Determine whether any outliers exist.
- Draw a boxplot to summarise the data, marking outliers if they exist.

a 4, 6, 5, 2, 3, 4, 4, 13, 8, 7, 6

b 1.8, 1.7, 1.8, 1.9, 1.6, 1.8, 2.0, 1.1, 1.4, 1.9, 2.2

c 21, 23, 18, 11, 16, 19, 24, 21, 23, 22, 20, 31, 26, 22

d 37, 48, 52, 51, 51, 42, 48, 47, 39, 41, 65

Outliers:
more than $Q_3 + 1.5 \times \text{IQR}$
or
less than $Q_1 - 1.5 \times \text{IQR}$.
Mark with a X.
The next value above or below an outlier is used as the new end of the whisker.



7 A butcher records the weight (in kilograms) of a dozen parcels of sausages sold on one morning.

1.6 1.9 2.0 2.0 2.1 2.2
2.2 2.4 2.5 2.7 3.8 3.9

a Write down the value of:

- | | | |
|------------------|------------------|------------------|
| i minimum | ii Q_1 | iii Q_2 |
| iv Q_3 | v maximum | vi IQR |

b Find any outliers.

c Draw a boxplot for the weight of the parcels of sausages.

Problem-solving and Reasoning



5E

- 8 Phillip the gardener records the number of days that it takes for 11 special bulbs to germinate. The results are:

8 14 15 15 16 16 16 17 19 19 24

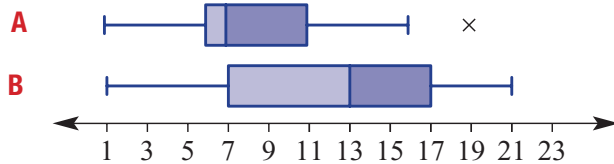
- a Write down the value of:

i minimum ii Q_1 iii Q_2 iv Q_3 v maximum vi IQR

- b Are there any outliers? If so, what are they?

- c Draw a boxplot for the number of days it takes for the bulbs to germinate.

- 9 Consider these parallel boxplots A and B.



Parallel boxplots are two boxplots that can be compared using the same scale.

Compare the boxplots at each point of the five-figure summary.

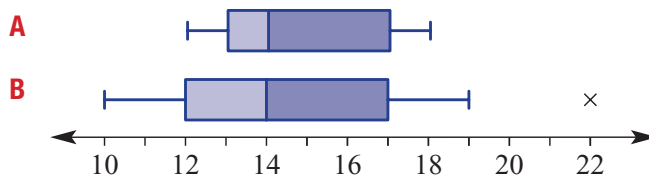
- a What statistical measure do these boxplots have in common?

- b Which data set (A or B) has a wider range of values?

- c Find the IQR for: i data set A ii data set B

- d How would you describe the main difference between the two sets of data from which the parallel boxplots have been drawn?

- 10 Two data sets can be compared using parallel boxplots on the same scale, as shown below.



- a What statistical measures do these boxplots have in common?

- b Which data set (A or B) has a wider range of values?

- c Find the IQR for:

i data set A ii data set B

- d How would you describe the main difference between the two sets of data from which the parallel boxplots have been drawn?



Creating parallel boxplots

- 11 Fifteen essays were marked for spelling errors by a particular examiner and the following numbers of spelling errors were counted.

3, 2, 4, 6, 8, 4, 6, 7, 6, 1, 7, 12, 7, 3, 8

The same 15 essays were marked for spelling errors by a second examiner and the following numbers of spelling errors were counted.

12, 7, 9, 11, 15, 5, 14, 16, 9, 11, 8, 13, 14, 15, 13

- a Draw parallel boxplots for the data.

- b Do you believe there was a major difference in the way the essays were marked by the two examiners? If yes, describe this difference.

Using technology 5.5: Using calculators to draw box plots

This activity is available on the companion website as a printable PDF.